DT SIMULINK MATLAB

Fitsync: A Digital Twin-Based AI System for Real-Time Heart Rate Monitoring

1. Introduction

Fitsync is an innovative approach that integrates multiple existing technologies to address a critical social issue—the increasing number of deaths during workouts. The growing trend of sudden cardiac arrests and overexertion incidents in fitness centers highlights the urgent need for self-monitoring solutions. While gyms and personal trainers offer valuable guidance, they lack a real-time monitoring system capable of assessing an individual's health based on past and present data.

Fitsync presents a unique AI-powered system that leverages machine learning (ML) models trained on fitness datasets to provide users with real-time insights into their heart rate (HR) conditions. The system enables users to visualize their live HR, predicted HR based on previous patterns, and anomaly detection results through an interactive web-based dashboard. The core technology underpinning Fitsync is the Digital Twin (DT), a cutting-edge approach with limited adoption in the fitness and medical sectors.

This document details the implementation of the Digital Twin in MATLAB Simulink, explaining each component in depth, including serial communication, preprocessing, ML model integration, anomaly detection, and real-time data visualization.

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2. Understanding Digital Twin and Its Role in Fitsync

A Digital Twin is a virtual representation of a physical system that continuously receives real-time data from its physical counterpart to simulate, analyze, and predict its behavior. In the case of Fitsync, the Digital Twin mirrors a user's physiological state by continuously processing heart rate data. The advantages of using a Digital Twin for Fitsync include:

- **Real-time monitoring:** Users receive instant feedback on their heart rate trends.
- **Predictive insights:** The ML model forecasts future HR values based on historical data.
- **Anomaly detection:** Deviations from expected HR patterns can be flagged for further analysis.
- **User engagement:** A web-based interface provides a comprehensive view of the user's workout statistics.

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3. Implementation of the Digital Twin in MATLAB Simulink

The Digital Twin for Fitsync is implemented in MATLAB Simulink, integrating hardware components (Arduino and BPM sensor) with software models for data processing and visualization.

3.1. Data Acquisition from Arduino

- **Serial Receive Block:** Reads HR data from an Arduino device over a USB serial connection.
- **Baud Rate Configuration:** Set at **115200** for high-speed data transfer.
- **Preprocessing:** The incoming raw data is processed for format conversion and noise reduction.

3.2. Data Preprocessing

- **Conversion Block:** Normalizes the HR data before feeding it into the ML model.
- **Reshape Data Block:** Reformats data for compatibility with the LSTM-based model.

3.3. Machine Learning Model Integration

- **ONNX Model Import:** The pre-trained LSTM Autoencoder model is imported into Simulink using the **Deep Learning Toolbox for ONNX Model Converter**.

- **Model Input & Output:** The model receives preprocessed HR data and outputs the predicted HR.
- **Normalization and Denormalization:** Ensures that data fed into the model and the output predictions maintain consistency with real-world values.

3.4. Anomaly Detection

- **Detect Anomaly Block:** Compares actual HR values with predicted HR to flag abnormal variations.
- **Threshold Definition:** A threshold value is used to distinguish between normal and abnormal HR patterns.

3.5. Real-Time Data Communication and Visualization

- **UDP Send Blocks:** Transmit HR and anomaly data to the web-based interface for user display.
- **Scope Blocks:** Display real-time HR signals and detected anomalies within Simulink.
- **Control Communication Toolbar:** Enables real-time tuning and monitoring of Simulink components.
- **Simulink Support Package for Arduino Hardware:** Facilitates seamless data exchange between MATLAB and the Arduino-based BPM sensor.

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4. Current Status and Future Enhancements

At present, the system flags anomalies on a graph but does not trigger alerts for users. Future developments will focus on:

- Implementing a real-time alert system to notify users of potential health risks.
- Enhancing model accuracy with more training data and advanced deep learning techniques.
- Expanding the web interface to include more interactive user features.

By leveraging Digital Twin technology, Fitsync provides an unprecedented level of real-time fitness monitoring, ensuring a safer and more data-driven workout experience for users.